DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

UNIVERSITY EXAMINATIONS 2021/2022 ACADEMIC YEAR

THIRD YEAR EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CIT 2207: MOBILE COMPUTING

DATE: 30/11/2021 **TIME: 02.00-04.00 P.M.**

Instructions: Answer Question 1 and Any Other Two

Question 1: (30 Marks)

a) Differentiate between *mobility* and *wireless* terminologies as used in mobile computing. (2 marks)

b) Explain the role of Wireless Markup Language (WML) in wireless applications. (2 marks)

c) Assume that your friend recently travelled to West Africa and you just made a phone call to her. She is surprised and is asking how her Safaricom Kenya line worked in Ghana. Discuss mobile IP topology and explain how it facilitates such kind mobility management beyond a particular location area (4 marks)

d) Using a sketch diagram discuss any four key components of a cellular wireless network architecture (4 marks)

e) Explain three reasons for a handoff to be conducted in mobility management (3 marks)

f) Compare the working of FDMA and TDMA schemes (4 marks)

g) Explain any TWO problems associated with Mobile IP (2 marks)

h) Describe the components of the wireless application protocol (WAP) architecture (5 marks)

i) Many a times we visit overcrowded places such as graduations ceremonies for university students or even when police officers are graduating we have had cases where by family, relative and friends of individual graduating fail to meet the graduand or even each other because during that time calls fails to go through, sometime calls terminates and fail to connect again. To reduce force termination and promote call completion channel assignment scheme have been proposed. Discuss four channel assignment schemes that have been proposed (4 marks)

Question 2: (20 Marks)

a) Describe in detail the following occurrence of media access in mobile computing. Use diagram where necessary.

i. Exposed Terminals (2 marks)

ii. Hidden Terminal (2 marks)

iii. Near and Far Terminal (2 marks)

b) Discuss four types of handover (handoff) managements used in mobile cellular technologies (8 marks)

c) John Travelled to France for a holiday. While in France John was able to check balance on his mobile phone, he could also receive calls and short messages as well as send and call his family back home in Kenya using his Safaricom Line. This can be explained using roaming management in mobility; thus, one is able to make calls or even receive messages while outside his or her geographical region. Using a diagram discuss the three key components of signaling system 7 (SS7) network that necessitated John to make and receive calls and messages while in France. (6 marks)

Question 3: (20 Marks)

a) Mobile communication is resource constrained. Therefore, it requires special operating environment which translates TCP/IP protocols into an equivalent of wireless applications environment. Using a model diagram, discuss how the WAP architecture seamlessly integrates www internet applications (6 marks)

b) Using a diagram, describe the IP mobile registration process indicating clearly all entities involved in registration process. (4 marks)

c) Briefly explains the improvements of mobile IPV6 that has solved the challenges raised in Mobile IPV4. (2 marks)

d) Explain the steps involved in the call delivery procedure in GSM network in the following cases: (6 marks)
   i. GSM mobile terminated call
   ii. GSM mobile originated call

e) Explain mobility binding in mobile ip process (2 marks)

Question 4: (20 Marks)

a) Define the following terms (3 marks)
   i. Session Mobility
   ii. Service Mobility
   iii. Network mobility

b) Explain how Frequency reuse help channel allocation of cellular systems (3 marks)

c) Explain FOUR challenges of mobile computing (4 marks)

d) You have been recruited as an advisor on information technology for a firm in the hotel industry. The firm intends to design and implement an online reservation system. The system will have the client and the server side. Your work is to design and implement the client side so that customers can reserve and book rooms online. The client side will have the following: -

   i. Heading – HILTON HOTEL RESERVATION ONLINE
   ii. Personal Information
      a. Name:
      b. Telephone Number
      c. Email

iii. Accommodation Requirements
    a. Room type
    b. Package
iv. Ages of children
v. Duration of stay
vi. Submit and Reset Button

The personal information should be contained in a textbox. (Name, telephone and email) The accommodation requirements should be contained in a drop menu (room type and package)
The ages of the children should be contained in a checkbox
Duration of stay should be contained in a radio box

Required

i. Using a sketch diagram design the above information (2 marks)
ii. Use the WML and form tags to implement (i) above (5 marks)
iii. Write WML script that will validate the user inputs before it submitted to the server (3 marks)

END.